Sham Satish Thakare

Independent Computer Science Researcher & Data Engineer / ML Specialist

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Research Statement & Interests

I conduct foundational and systems research on **Trustworthy, Adaptive, and Verifiable Autonomous Systems**. My work focuses on four primary vectors:

[1] **Foundation Model Plasticity & Calibrated Reasoning**: Representation geometry diagnostics ($\alpha_{\text{SVD}}, \bar{H}, \sigma_g^2$) for predicting reinforcement learning reward plasticity in intermediate checkpoints, token entropy length confounding, and RLVR self-consistency calibration.

[2] **Confidential Computing & Hardware Security**: Zero-knowledge remote attestation membership proofs and access-frequency-weighted adaptive Oblivious RAM (ORAM) tree rebalancing algorithms for hardware enclaves (SGX/TDX).

[3] **Learning-Augmented Distributed Consensus**: Failure-domain aware dynamic quorum adaptation (AdaptiveReplica) over Raft joint-consensus transitions with uncertainty-aware trust gates (T_3).

[4] **Causal Observability & Verifiable AI Systems**: Topological causal walks over OpenTelemetry microservice dependency graphs (TraceMind) and zero-knowledge authorization-path provenance graph proofs (EnclaveShield).

Education

Anantrao Pawar College of Engineering & Research (APCOER), *Pune, India* **Jan 2020 – Jun 2024**
Bachelor of Technology in Artificial Intelligence and Data Science

- **Cumulative GPA**: 8.70 / 10.00
- **Academic Honors**: Awarded 100% Merit Scholarship for 8 consecutive semesters based on academic performance.
- **Core Coursework**: Engineering Mathematics, Scientific Computing, Operating Systems, Design and Analysis of Algorithms, C++, Python, Data Structures, Computer Networks, Database Management Systems, Artificial Intelligence, Machine Learning, Computational Modeling, Signal Processing, Linear Algebra, Probability and Statistics.

Research Experience & Manuscripts

Submitted / Under Review Manuscripts

[1] **When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in LM RL**
Sham Satish Thakare
IEEE Transactions on Artificial Intelligence (IEEE TAI) [**Submitted, Aug 2026**]
Exposes length confounding in token predictive entropy ($r = +0.486$), proving sample-level uncertainty weighting (CA-GRPO) yields 0.00% Pass@1 gain over standard outcome GRPO. Code: `https://github.com/shamddd/ear_grpo_reasoning`

[2] **recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning**
Sham Satish Thakare
11th IEEE Special Session on Machine Learning on Big Data (MLBD 2026) / IEEE BigData [**Submitted (BigD497), Aug 2026**]
Introduces a matched state-contrast protocol ($D_{\text{recovery}} = -0.1100$, 95% CI [-0.240, +0.030]), demonstrating Instruct checkpoints lack a recovery-specific advantage over Base checkpoints on single-step arithmetic errors. Code: `https://github.com/shamddd/recovery_eval`

[3] **StateShift: State-Matched Evaluation of Error Recovery in Neural Reasoning**
Sham Satish Thakare
Artificial Intelligence (Elsevier AIJ) [**Submitted (ARTINT-D-26-01491), 2026**]
Formulates verifier-defined continuation state matching on neural reasoning rollouts under structural covariate controls. Code: `https://github.com/shamddd/stateshift`

Preprints & Working Papers

[4] **Amortized Intervention Frontiers for Language-Model Reasoning: When Does Training Beat Search?**
Sham Satish Thakare

Working Paper (Target under consideration: TMLR) [**Working Paper, 2026**]
Formalizes deployment cost frontiers ($C_{\text{total}} = C_{\text{train}} + Q \cdot C_{\text{inference}}$). Demonstrates OOD length extrapolation accelerates RLVR amortization crossover ($R_f \approx 0.0618 \ll 1.0$). Code: https://github.com/shamddd/submission_package

[5] **Predicting Reinforcement-Learning Plasticity of Intermediate Language-Model Checkpoints: A Cross-Architecture Diagnostic Study**
Sham Satish Thakare
Working Paper (Target under consideration: JMLR) [**Working Paper, 2026**]
Formulated diagnostic probing vectors $\phi(C_k)$ measuring layer-wise representation entropy and singular value spectrum decay rates ($R^2 = 0.91, p = 0.0004$). Code: https://github.com/shamddd/adaptive-rl-forge

[6] **EnclaveShield: Zero-Knowledge Memory Attestation and Side-Channel Mitigation for Hardware Enclaves**
Sham Satish Thakare
Working Paper (Target under consideration: IEEE TDSC) [**Working Paper, 2026**]
Engineered ZK quote attestation membership proofs and access-frequency-weighted adaptive Path ORAM tree rebalancing (1.47ms latency, $H(A) = 0.82 \pm 0.02$). Code: https://github.com/shamddd/enclaveshield

[7] **AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus**
Sham Satish Thakare
Working Paper (Target under consideration: IEEE TPDS) [**Working Paper, 2026**]
Formulated dynamic vote-weight adaptation over Raft joint-consensus transitions with trust gates (T_3), reducing p99 write latency to 13.50ms (88.8% reduction vs static 120.48ms) with zero stale reads. Code: https://github.com/shamddd/quorumshift

[8] **TraceMind: Graph-Constrained Causal Reasoning for Root-Cause Localization in Microservice Systems**
Sham Satish Thakare
Working Paper (Target under consideration: IEEE TCC) [**Working Paper, 2026**]
Constructed topological causal walks over OpenTelemetry Service Dependency Graphs, achieving 100.0% Top-1 RCA accuracy (MRR = 1.00) across 24 fault scenarios. Code: https://github.com/shamddd/tracemind

Professional Work Experience

Spectra Corporate Service, *Pune, India* **Jun 2024 – Present**
Data Engineer / Machine Learning Specialist

- Architected scalable data pipelines using C++, Python, and SQL on GCP and AWS (GKE, Compute Engine, Cloud Storage, Cloud Logging).
- Implemented Linux multithreaded system software, synchronization primitives, container security controls, and IaC pipelines using Terraform, Docker, Kubernetes, and GitHub Actions.
- Constructed Prometheus and Grafana distributed telemetry dashboards for SLA monitoring and proactive incident detection.

APCOER Pune, *Pune, India* **Jun 2023 – Dec 2024**
Research Intern / Teaching Fellow

- Served as Teaching Fellow for *Intro to Algorithms and Their Limitations* and *Foundations of ML: AI Alignment and Safety* (Aug–Dec 2024).
- Served as Course Producer for *Fundamentals of Computation*, *Introduction to Algorithms*, and *Theory of Computation* (Aug 2022–May 2023).
- Assisted faculty in experimental design, statistical analysis, and open-source software reproducibility protocols.

Technical Skills

- **Languages**: Python, C++, C, Rust, Go, Java, SQL, Bash, LaTeX
- **AI & Machine Learning**: PyTorch, GRPO/PPO RL Probes, Transformers, HuggingFace, NumPy, SciPy, Scikit-Learn
- **Systems & Security**: Raft Consensus, OpenTelemetry, TEEs (SGX/TDX), Path ORAM, Zero-Knowledge Proofs
- **Cloud & Infrastructure**: Docker, Kubernetes, Helm, Terraform, GCP, AWS, Prometheus, Grafana, Linux Kernel/POSIX